Sigrok BMS Protocol Decoder Capture Analysis

Generated from offline sigrok/PulseView analysis outputs

Executive Summary

30	**17**	**13**	**19 / 6 / 5**
total captures	RS485/UART captures	CAN captures	Bridge / Forward / Direct

Key observations

- Anenji Pylon RS485 JKBMS shows the clearest RS485 latency change: Direct cable is -4581.73 (-47.2%) versus Bridge for request-to-response average.
- Growatt RS485 JKBMS has a much longer Direct cable tail latency: Direct cable is +21480.84 (+40.1%) versus Bridge for full-exchange P95.
- Growatt CAN SeplosBMS cycles are shorter in Direct cable mode: Direct cable is -164247.78 (-19.3%) versus Bridge for cycle average.
- Growatt CAN JKBMS frame rate stays effectively flat across modes: Direct cable is +0.1 (+0.2%) versus Bridge.

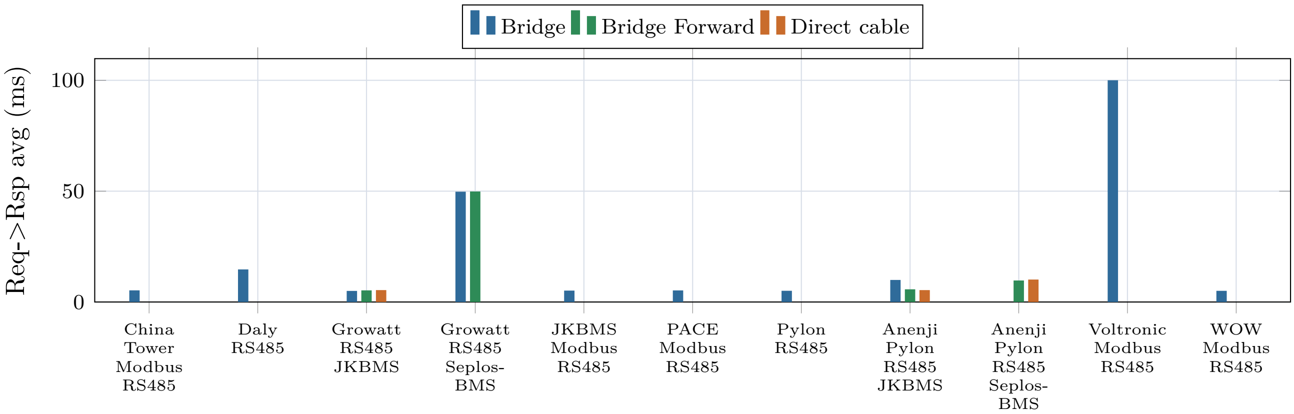
Table 1: Largest Direct cable versus Bridge changes by absolute percentage.

Group	Metric	Direct vs Bridge	Abs. change
Growatt CAN JKBMS	Cycles/s	+0.1 (+100%)	100%
Growatt CAN JKBMS	Cycle min (ms)	-5000.47 (-50.1%)	50.1%
Growatt CAN JKBMS	Cycle avg (ms)	-5000.47 (-50.1%)	50.1%
Growatt CAN JKBMS	Cycle max (ms)	-5000.47 (-50.1%)	50.1%
Growatt CAN JKBMS	Cycle P95 (ms)	-5000.47 (-50.1%)	50.1%
Growatt CAN SeplosBMS	Cycle min (ms)	-121.3 (-48.3%)	48.3%

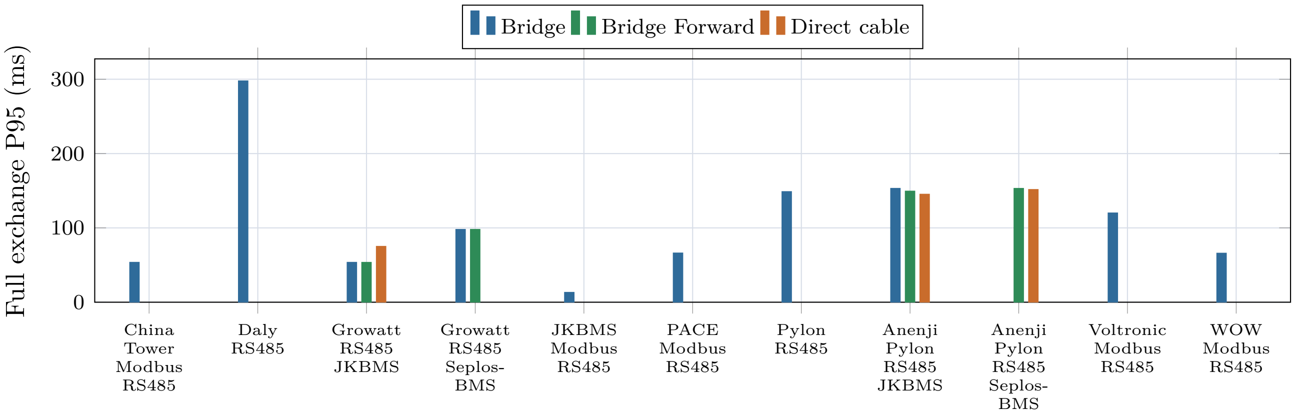
1 Protocol Overview Charts

These charts include every protocol group available in the overview CSV. Missing topology modes are intentionally left blank, so a group can have one, two, or three bars depending on the captures currently available.

CAN cycle duration charts use a logarithmic y-axis because values span multiple orders of magnitude, from sub-millisecond single-frame cycles to multi-second capture-length cycles.

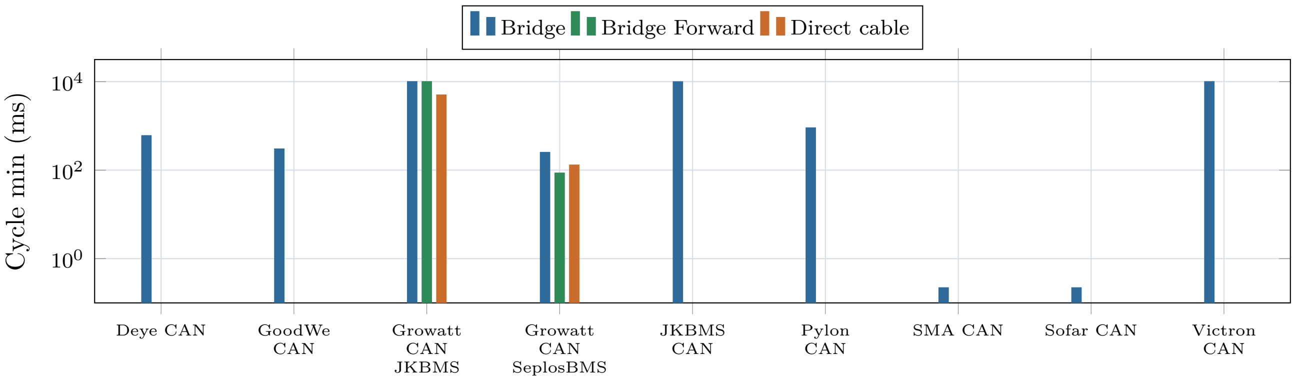


(a) Average request-to-response latency for all available RS485/UART protocol groups.

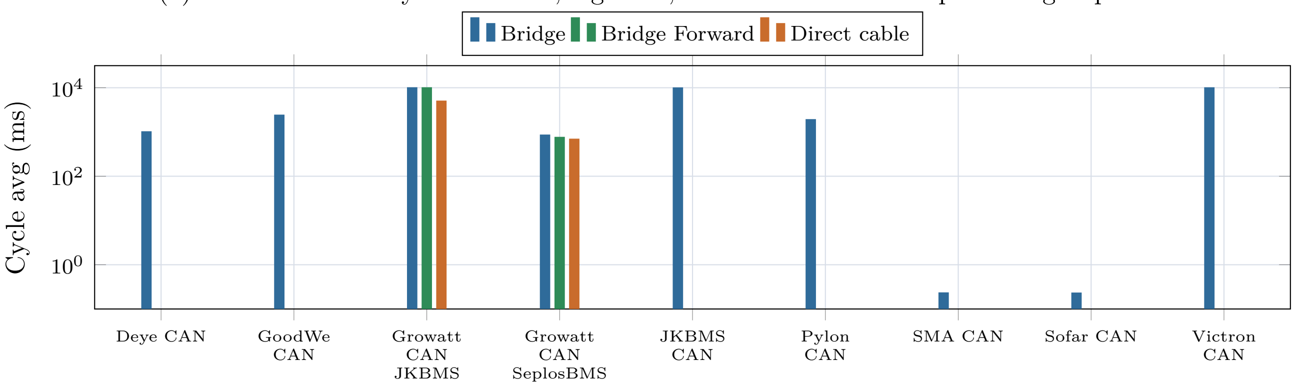


(b) P95 full-exchange duration for all available RS485/UART protocol groups.

Figure 1: RS485/UART protocol overview charts.

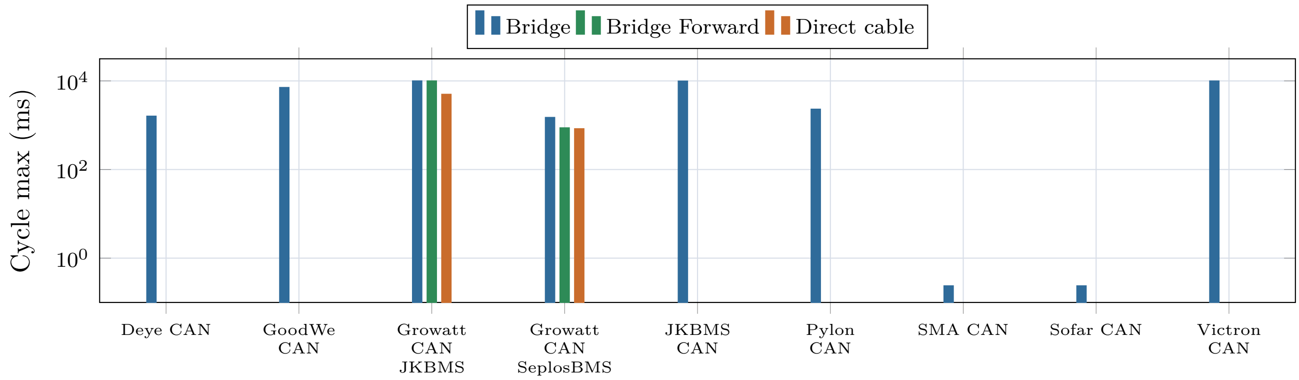


(a) Minimum CAN cycle duration, log scale, for all available CAN protocol groups.

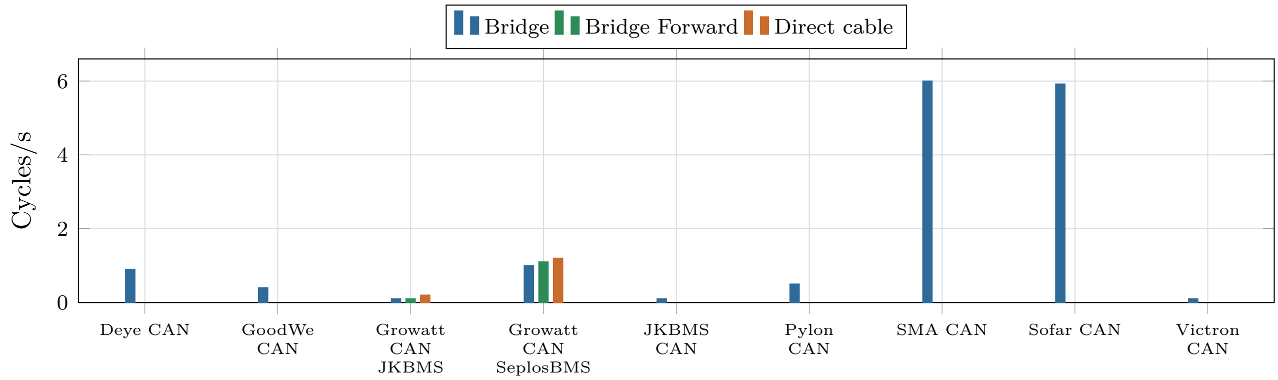


(b) Average CAN cycle duration, log scale, for all available CAN protocol groups.

Figure 2: CAN cycle duration overview charts.

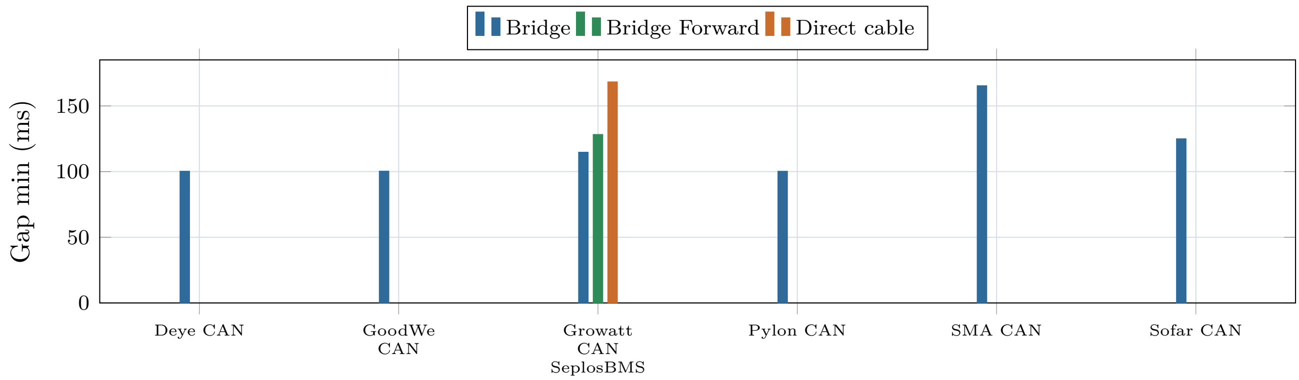


(a) Maximum CAN cycle duration, log scale, for all available CAN protocol groups.

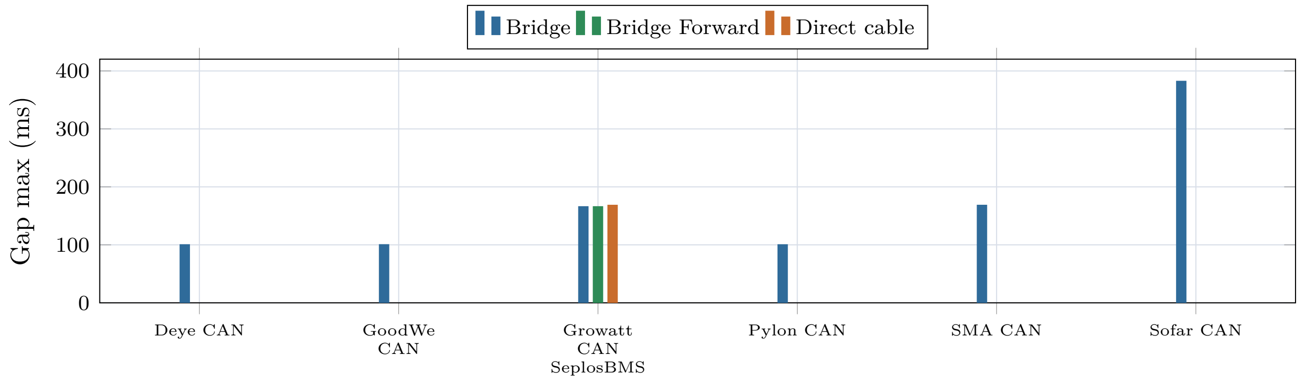


(b) CAN cycle rate normalized by capture duration for all available CAN protocol groups.

Figure 3: CAN maximum cycle duration and throughput charts.



(a) Minimum inter-cycle gap for CAN protocol groups with at least two detected cycles.



(b) Maximum inter-cycle gap for CAN protocol groups with at least two detected cycles.

Figure 4: CAN inter-cycle gap overview charts.

2 Three-mode Delta Tables

The following tables focus only on groups where all three modes are available: Bridge, Bridge Forward, and Direct cable. Deltas are directional: orange means an increase, green means a decrease. Whether an increase is good depends on the metric.

2.1 RS485/UART Three-mode Deltas

Group	Metric	Bridge	Forward	Direct	Forward vs Bridge	Direct vs Bridge	Direct vs Forward
Growatt RS485 JKBMS	Complete exchanges/s	2.5	2.5	2	0 (0%)	-0.5 (-20%)	-0.5 (-20%)
Growatt RS485 JKBMS	Req->Rsp avg (us)	4760	5008.1	5126.76	+248.1 (+5.2%)	+366.76 (+7.7%)	+118.66 (+2.4%)
Growatt RS485 JKBMS	Req->Rsp P95 (us)	5294.32	5293.78	5438.65	-0.54 (0%)	+144.33 (+2.7%)	+144.87 (+2.7%)
Growatt RS485 JKBMS	Full exchange avg (us)	49094.79	48926.21	59337.41	-168.58 (-0.3%)	+10242.62 (+20.9%)	+10411.2 (+21.3%)
Growatt RS485 JKBMS	Full exchange P95 (us)	53520.07	53484.77	75000.91	-35.3 (-0.1%)	+21480.84 (+40.1%)	+21516.13 (+40.2%)
Anenji Pylon RS485 JKBMS	Complete exchanges/s	0.8	0.8	0.8	0 (0%)	0 (0%)	0 (0%)
Anenji Pylon RS485 JKBMS	Req->Rsp avg (us)	9706.67	5488.62	5124.94	-4218.05 (-43.5%)	-4581.73 (-47.2%)	-363.68 (-6.6%)
Anenji Pylon RS485 JKBMS	Req->Rsp P95 (us)	9912.79	6190.08	5692.96	-3722.71 (-37.6%)	-4219.84 (-42.6%)	-497.12 (-8%)
Anenji Pylon RS485 JKBMS	Full exchange avg (us)	111164.32	106950.36	102917.89	-4213.95 (-3.8%)	-8246.42 (-7.4%)	-4032.47 (-3.8%)
Anenji Pylon RS485 JKBMS	Full exchange P95 (us)	153034.21	149320.43	145154.53	-3713.79 (-2.4%)	-7879.68 (-5.1%)	-4165.9 (-2.8%)

2.2 CAN Three-mode Deltas

Group	Metric	Bridge	Forward	Direct	Forward vs Bridge	Direct vs Bridge	Direct vs Forward
Growatt CAN JKBMS	Frames/s	51.9	52	52	+0.1 (+0.2%)	+0.1 (+0.2%)	0 (0%)
Growatt CAN JKBMS	Cycles/s	0.1	0.1	0.2	0 (0%)	+0.1 (+100%)	+0.1 (+100%)
Growatt CAN JKBMS	Cycle min (us)	9980699.27	9980680.5	4980232.08	-18.77 (0%)	-5000467.18 (-50.1%)	-5000448.42 (-50.1%)
Growatt CAN JKBMS	Cycle avg (us)	9980699.27	9980680.5	4980232.08	-18.77 (0%)	-5000467.18 (-50.1%)	-5000448.42 (-50.1%)
Growatt CAN JKBMS	Cycle max (us)	9980699.27	9980680.5	4980232.08	-18.77 (0%)	-5000467.18 (-50.1%)	-5000448.42 (-50.1%)
Growatt CAN JKBMS	Cycle P95 (us)	9980699.27	9980680.5	4980232.08	-18.77 (0%)	-5000467.18 (-50.1%)	-5000448.42 (-50.1%)
Growatt CAN Seplos-BMS	Frames/s	13	12.9	13	-0.1 (-0.8%)	0 (0%)	+0.1 (+0.8%)
Growatt CAN Seplos-BMS	Cycles/s	1	1.1	1.2	+0.1 (+10%)	+0.2 (+20%)	+0.1 (+9.1%)
Growatt CAN Seplos-BMS	Cycle min (us)	251234.86	85691.12	129932.96	-165543.74 (-65.9%)	-121301.9 (-48.3%)	+44241.84 (+51.6%)
Growatt CAN Seplos-BMS	Cycle avg (us)	851373.72	755942	687125.94	-95431.72 (-11.2%)	-164247.78 (-19.3%)	-68816.06 (-9.1%)
Growatt CAN Seplos-BMS	Cycle max (us)	1496223.99	871814.91	831769.22	-624409.08 (-41.7%)	-664454.76 (-44.4%)	-40045.68 (-4.6%)
Growatt CAN Seplos-BMS	Cycle P95 (us)	1221318.48	861952.16	831768.79	-359366.32 (-29.4%)	-389549.69 (-31.9%)	-30183.37 (-3.5%)
Growatt CAN Seplos-BMS	Inter-cycle gap min (us)	114668.76	128174.72	168225.68	+13505.97 (+11.8%)	+53556.93 (+46.7%)	+40050.96 (+31.2%)
Growatt CAN Seplos-BMS	Inter-cycle gap avg (us)	155493.47	160226.81	168228.48	+4733.34 (+3%)	+12735.01 (+8.2%)	+8001.67 (+5%)
Growatt CAN Seplos-BMS	Inter-cycle gap max (us)	165777.83	165777.83	168229.68	0 (0%)	+2451.84 (+1.5%)	+2451.84 (+1.5%)
Growatt CAN Seplos-BMS	Inter-cycle gap P95 (us)	165776.23	165777.83	168229.68	+1.6 (0%)	+2453.45 (+1.5%)	+2451.84 (+1.5%)

3 Capture Overview

3.1 RS485/UART Capture Overview

Target	Topology	Frames	Complete	Incomplete	Req->Rsp avg (us)	Req->Rsp P95 (us)	Full P95 (us)
china tower modbus	Bridge	79	39	1	5015.03	5337.93	53525.72
daly rs485	Bridge	88	44	0	14458.98	18060.3	297646.8
growatt rs485	Bridge	50	25	0	4760	5294.32	53520.07
growatt seplos rs485	Bridge	49	24	1	49517.09	49863.54	97806.11
direct growatt rs485	Direct cable	100	50	0	5126.76	5438.65	75000.91
forward growatt rs485	Bridge Forward	50	25	0	5008.1	5293.78	53484.77
forward growatt seplos rs485	Bridge Forward	49	24	1	49604.18	50490.32	97846.4
jkbms modbus	Bridge	78	39	0	4903.5	5191.55	13115.59
pace modbus	Bridge	79	39	1	4987.74	5260.88	66064.78
pylon rs485	Bridge	49	18	5	4817.5	5537.46	148667.74
anenji pylon rs485	Bridge	20	8	4	9706.67	9912.79	153034.21
direct anenji jkbms pylon rs485	Direct cable	61	20	21	5124.94	5692.96	145154.53
direct anenji seplos pylon rs485	Direct cable	50	20	10	9887.21	12043.43	151498.7
forward anenji pylon rs485	Bridge Forward	24	8	8	5488.62	6190.08	149320.43
forward anenji seplos pylon rs485	Bridge Forward	20	8	4	9469.73	9877.58	153001.64
voltronic modbus	Bridge	78	39	0	99796.75	101837.19	119962.1
wow modbus	Bridge	78	39	0	4799.13	5093.6	65808.75

3.2 CAN Capture Overview

Target	Topology	Frames	IDs	Cycles	Cycle min (us)	Cycle avg (us)	Cycle max (us)	Cycle P95 (us)	Errors
deye can	Bridge	100	8	9	599848.77	1011013.33	1599872.42	1599871.23	0
goodwe can	Bridge	100	8	4	300245.9	2399973.05	7100062.15	6275024.24	0
growatt can	Bridge	519	10	1	9980699.27	9980699.27	9980699.27	9980699.27	0
growatt seplos can	Bridge	130	12	10	251234.86	851373.72	1496223.99	1221318.48	0
direct growatt can	Direct cable	260	10	1	4980232.08	4980232.08	4980232.08	4980232.08	0
direct growatt seplos can	Direct cable	65	12	6	129932.96	687125.94	831769.22	831768.79	0
forward growatt can	Bridge Forward	520	10	1	9980680.5	9980680.5	9980680.5	9980680.5	0
forward growatt seplos can	Bridge Forward	129	12	11	85691.12	755942	871814.91	861952.16	0
jkbms can	Bridge	344	12	1	9907992.71	9907992.71	9907992.71	9907992.71	0
pylon can	Bridge	100	8	5	900275.94	1900011.26	2299947.27	2299945.2	0
sma can	Bridge	60	6	60	219.4	230.73	237.4	237.4	0
sofar can	Bridge	148	6	148	219.4	228.76	237.4	237.4	0
victron can	Bridge	250	19	1	9960725.84	9960725.84	9960725.84	9960725.84	0

Notes

- The LaTeX source is generated from committed CSV files under `analysis/out/`.
- Capture durations are not identical, so normalized rates are included where useful.
- CAN cycle duration can be dominated by capture length when only one cycle is detected.
- Compiled PDFs and LaTeX build artifacts should stay out of the repository.